

Three Addresses of a Connectome-Constrained Mushroom-Body Memory: Dopamine Chooses the Compartment, Sparse Kenyon-Cell Codes Set the Cue, Side- and Subtype-Specific Routing Shapes the Output

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Abstract—We asked which constraints of the mushroom-body (MB) wiring determine where a dopamine-gated memory is stored, what retrieves it, how long it lasts and how it reaches the rest of the brain, using a leaky integrate-and-fire model of the adult male *Drosophila* CNS connectome (166,700 neurons) with local plasticity on its 61,210 Kenyon-cell (KC) to MB-output-neuron (MBON) synapses. State transplants show that, from about one second after a write, the memory is carried only by the KC→MBON weights, and its lifetime is exactly the prescribed synaptic decay. Retrieval depends on input sparseness: with dense input (512 of 675 projection neurons per symbol) 87% of KCs respond to every probe and the MB acts as one scalar store per MBON; with sparse input (160–192) the written item reads out 3.6–4.7 times more strongly than other probes, and scrambling which KCs carry the change removes 34–89% of that readout. Random projection-neuron-to-KC wiring suffices for this cue. What distinguishes different items at the output needs the real wiring, but only its coarse organisation: randomising either wiring layer makes item outputs nearly identical (cosine 0.76 against 0.96–0.99), whereas swapping KCs within a subtype and body side does not. When different dopaminergic neurons teach, the teacher decides the compartment (87% of output variance, against 5% for the item). Interference between memories is linear and set by KC overlap, and downstream neurons receive a memory only when it changes an MBON spike (35 of 35 against 0 of 93 cells).

I. INTRODUCTION

This study grew out of an attempt to use the connectome as a frozen reservoir for character-level language modelling on TinyShakespeare [21]. Adding local KC→MBON plasticity to that model wrote information into the synaptic state that could be decoded, yet the language readout did not improve. Those experiments were run on a simulator version later found to misplace plastic writes (Section IV), and their language scores turned out to be carried largely by an explicit hashed character-context head; but the question they raised, why information written into the MB is not read out, is the starting point here. The companion report [21] audits that history and uses the mechanisms found below to revise the language model: with sparse input, the brain’s features improved a hashed context-only model in all 18 held-out cases with 20k training

characters, but not a smoothed 5-gram, and the gain vanished with more data; there the frozen brain acts as a fading memory of about four characters, whatever its wiring.

The MB stores associative memories as dopamine-gated, compartment-specific changes of KC→MBON synapses [1], [2], [3]. Its anatomy suggests a division of labour: an expansion from about 50 glomerular channels onto thousands of sparsely active KCs [4], [5], apparently random at the level of PN→KC convergence [6], followed by a compartmental KC→MBON layer with dopaminergic teaching signals [2], [7], [8]. Theory argues that such expansion and connectivity degree are chosen for what the downstream plasticity must do [9].

Much of this is established. Sparse, decorrelated KC codes improve learned odour discrimination and have been proposed to provide precisely addressable locations for odour-specific memories [5]. Models of the MB have implemented dopamine-gated KC→MBON plasticity and reinforcement learning [14], the integration of stochastic KC input by a single MBON has been modelled from its EM architecture [15], and a connectome-constrained model combined with voltage imaging has explained dopamine-mediated interactions between short- and long-term memory units [16]. Whole-brain connectomes have been turned into executable models of sensorimotor processing [10], [11] and perturbed in silico to find the elements that sustain spontaneous activity [17].

What has not been done is to take one executable whole-CNS model with local plasticity and decompose a memory causally into its parts: which state variable carries it after activity has decayed, which wiring determines which cue retrieves it, which wiring determines what it evokes at the output, which part of the dopaminergic system decides where it is written, what sets its lifetime, when two memories interfere, and when a stored trace becomes visible downstream. We do this in a model of the complete male CNS connectome [13], answering each question with an intervention, a matched control and, where possible, a restoration: state transplants, weight scrambles that preserve the multiset of weight changes,

topology-preserving wiring swaps, teacher swaps and decay controls. We do not claim a memory capacity, and none of the components (sparse coding, dopamine-gated plasticity, executable connectomes) is new; the contribution is the decomposition and the conditions under which each component does or does not operate.

An earlier analysis of this model [22] reported that plasticity bought no readout advantage and that the write-effect direction was fixed by the probe rather than the content. Section IV explains why those conclusions do not stand: part of their data came from a simulator version that wrote to the wrong synapses, two of their confirmed predictions hold by construction, and all of their experiments ran at a dense input operating point in which the MB’s sparse code is absent.

II. MODEL AND METHODS

A. Connectome and dynamics

The substrate is the adult male *Drosophila* CNS connectome (MaleCNS v1.0, synapse confidence threshold 0.5) [13], simulated with the *flybrain* leaky integrate-and-fire package, with 166,700 neurons and 25,582,938 synaptic entries, of which 25,088,107 remain after removing inputs onto sensory neurons. Populations: 4,064 KCs, 97 MBONs, 356 dopaminergic neurons (PAM/PPL/PPM), 675 projection neurons (PNs) in 180 annotated types, and two APL neurons. Neurons are leaky integrate-and-fire units ($\Delta t = 20$ ms, $\tau = 100$ ms, threshold 1, reset 0) driven by $v \leftarrow e^{-\Delta t/\tau}v + gWz + b$, with gain $g = 1.5$, tonic input $b = 0.05$ and no intrinsic noise; only spikes z propagate. APL inhibits KCs (summed APL→KC weight ≈ -190 over $\sim 2,300$ edges per APL) and fires 1–2 times per input token at the legacy operating point.

B. Plasticity

Only the 61,210 existing KC→MBON entries are plastic: $w_{km} = w_{km}^0(1 + m_{km})$, with $m_{km} \leftarrow e^{-T/\tau_w}m_{km} + \eta\gamma_m a_k$ per token of length $T = 120$ ms, clipped to $|m| \leq 0.9$. a_k is the KC’s spike count in the token divided by six, and γ_m is a per-MBON gate computed from the simulated dopaminergic spikes through the connectome’s own DAN→MBON edges with hand-set valence signs (PAM +1, PPL −1, PPM +0.5). $\eta = 0.02$ and $\tau_w = 16$ s unless stated. The rule, the signs and the decay are modelling choices; results that depend on them are reported as interactions between these choices and the wiring, not as properties of the wiring alone.

C. Protocol

Symbols are encoded as fixed random subsets of PNs (the *PN code*); each token injects the code for six steps (drive d on the first step, $0.5d$ after). A write presents one symbol for 32 tokens with a dopaminergic pulse (PAM, +2.0) on each token; a gap of G silent tokens follows with plasticity off and passive decay on; a probe token (drive $2.5d$) is read over six further silent steps. Readouts are (i) pre-reset membrane voltages of the 97 MBONs and of 256 central neurons with the largest MBON input, averaged per step over the settle window, (ii) the write-attributable MBON input current $I_m =$

TABLE I
MODEL AND PROTOCOL PARAMETERS (DEFAULTS; DEVIATIONS ARE STATED WHERE USED).

Parameter	Value
Time step / membrane τ	20 ms / 100 ms
Threshold / reset / refractory	1 / 0 / none
Gain g / tonic b / noise	1.5 / 0.05 / 0 (deterministic)
Token	6 steps (120 ms); drive d on step 1, $0.5d$ on steps 2–6
Symbol code	fixed random set of PNs; only the current symbol injected
PN code size	512 of 675 (legacy), 160–192 (sparse); drive scale 1.5
Plastic edges	61,210 existing KC→MBON entries
Learning rate η / clip	0.02 / $ m \leq 0.9$
Weight decay τ_w	16 s (4, 64 s and ∞ in Results B)
Eligibility trace	$\tau = 0.5$ s, not used in the update
DAN gate valence	PAM +1, PPL −1, PPM +0.5 (hand-set)
Teacher	+2.0 on all PAM, first step of each write token
Write / gap	32 tokens / G silent tokens (plasticity off)
Probe	drive $2.5d$, then 6 silent settle steps
Readout	MBON (97) and 256 central MBON targets, settle window
Replay floor	$\leq 2.4 \times 10^{-7}$ (MBON voltage)

$\sum_k \Delta w_{km} c_k / 6$, with c_k the probe’s KC spike counts, which isolates the synaptic effect from the spiking nonlinearity, and (iii) MBON spike rasters. The *operating point* is the PN code size n_{PN} (of 675) and the drive scale; 512 at scale 1.0 is the legacy setting of earlier work. Drive saturates above scale 1.5 because PN spiking is binary.

All parameters are collected in Table I.

D. Snapshot, restore and transplant

The complete dynamical state is the voltage vector, the last spike set (the only transmission delay), the harness spike traces, the KC eligibility trace (inert when, as here, it does not enter the update) and the synaptic state (m, w); all other weights are hashed and verified constant. Restoring a snapshot reproduced natural continuation to the level of the GPU’s own replay-to-replay variability (maximum absolute voltage difference 2.4×10^{-7} at MBONs against 2.4×10^{-7} between natural replays), while three injected errors were detected: $+10^{-3}$ on all MBON voltages ($\sim 1000\times$ the floor), restoring m without the matching w ($\sim 3 \times 10^6\times$), and $+10^{-3}$ on a single synapse ($4\text{--}5\times$).

E. Interventions and statistics

Weight scrambles permute the actual weight changes Δw (not m) across edges, preserving their multiset exactly, either within each MBON (keeping each MBON’s change distribution, destroying KC identity) or within each KC (keeping which KCs changed, destroying which MBON they address). Wiring nulls rewire PN→KC or KC→MBON edges while preserving in-degrees, out-degrees and weight multisets, without duplicate edges. The simulation is deterministic at zero noise; the units of replication are write pairs, items and null draws, all within one connectome, so we report ranges and medians

TABLE II

TRANSPLANT DECOMPOSITION AT THE LEGACY OPERATING POINT (MBON LEVEL, PAIRS a/c AND b/d , PROBES = BOTH ITEMS AND A NEUTRAL CUE). D_m AND D_s : RESPONSE CHANGE WHEN ONLY m OR ONLY s IS SWAPPED, RELATIVE TO THE NATURAL CONTENT DIFFERENCE.

Gap	Fast-state $\ \Delta v\ $	D_m/D_{nat}	D_s/D_{nat}
0	28–31	0.29–0.42	0.97–1.00
2 tokens	2.6–3.1	0.49–0.99	0.13–0.94
8 tokens	$\approx 2 \times 10^{-3}$	1.000	≤ 0.007
32 tokens	$\approx 7 \times 10^{-6}$	1.000	0.000

and do not attach p values to population claims. Decision rules were written into each script before its data were inspected.

III. RESULTS

A. After one second the memory is carried by the $KC \rightarrow MBON$ weights

We saved full states at the pre-probe instant of episodes that wrote item X or Y and transplanted the synaptic state m and the fast state s independently (Table II). With no gap, the probe response is dominated by residual activity. From eight tokens (0.96 s) on, swapping m reproduces the entire content difference and swapping s reproduces none of it, and the same holds on a common no-write fast state. The fast-state difference between episodes decays with $\tau \approx 0.1$ s, the membrane time constant, reaching float32 resolution by 3.84 s; no spikes remain two tokens after the write. At this operating point the recurrent wiring does not prolong activity-based memory. The same decomposition at sparse input (192 PNs) gives the same result: from eight tokens on, swapping m reproduces 99.9–100% of the content difference and swapping s at most 0.6% (Fig. 1).

B. Retention is the prescribed decay; the readout is thresholded

The synaptic difference between two writes follows e^{-t/τ_w} exactly for $\tau_w \in \{4, 16, 64\}$ s. With the decay removed, the write effect and the content difference at the readout stay constant for 123 s: the network’s activity contributes neither forgetting nor consolidation, because it is silent between write and probe. The readout, however, is not a smooth image of the trace. For $\tau_w = 16$ s the content difference of one probe stays above 98% of its initial value for 15 s and drops to 4% by 31 s, while 16% of the synaptic difference remains; other probes show steps and even transient increases. Effective readout time constants were 12–15 s for $\tau_w = 16$ s, shorter than the trace. The steps coincide with MBON spikes appearing or disappearing (Section III-H). A retention of 38.3% after 15.36 s, reported earlier as a model memory lifetime, equals $e^{-15.36/16}$, the prescribed constant.

The sparse regime makes the mechanism explicit (Fig. 2). The cued input current follows $e^{-t/16\text{s}}$ exactly. The write adds one MBON spike to the probe response (6 against 5); while that spike persists, the MBON voltage difference and the downstream difference stay within 7% of their initial values, and when it disappears, between 15.4 and 30.7 s, the

TABLE III

OPERATING-POINT SWEEP (DRIVE SCALE 1.5 UNLESS NOTED). KC FRACTIONS DURING WRITE TOKENS 2–32 AND AT THE PROBE; JACCARD BETWEEN SYMBOLS AT THE PROBE; KCs WRITTEN BY ONE 32-TOKEN WRITE.

n_{PN}	KC write	KC probe	Jaccard	KCs written
160	0.02–0.03	0.09	0.08	479
192	0.04–0.08	0.16	0.14	1,105
224	0.05–0.09	0.21	0.19	557
256	0.10–0.24	0.35	0.30	1,636
320	0.22–0.42	0.65	0.67	3,277
448	0.70–0.80	0.83	0.91	3,578
512 (1.0)	0.51–0.68	0.87	0.93	3,312

downstream difference drops to the numerical floor although 16% of the synaptic difference remains. With the decay removed, the extra spike and the downstream difference persist unchanged for 123 s. Probes that never flip an MBON spike leave the downstream response at the floor throughout.

C. The legacy operating point has no sparse KC code

At $n_{PN} = 512$, 51–68% of KCs spike in each write token and 87% during the probe; KC sets of different symbols overlap with Jaccard 0.55–0.67 during writing and 0.93 during the probe. APL fires but does not sparsen this input. Every probe therefore reads almost the same KCs, and the written trace is accessed as a per-MBON sum. Scrambling KC identity within each MBON leaves the readout current almost unchanged (83–88% of its magnitude, cosine 0.99–1.00), and a neutral cue reads the stored content as strongly as the written item (cue specificity 0.96–1.13).

Sweeping the PN code size (Table III) reveals a sparse regime between 160 and 192 PNs in which probes activate 9–16% of KCs, symbols share few KCs, and the write still changes MBON voltages by orders of magnitude above the numerical floor (Fig. 3). Below about 128 PNs the KC layer falls silent.

D. In the sparse regime, KC identity is the retrieval cue

Across six write pairs, the written item read the stored change through a current 3.6 (192 PNs; range 2.9–6.2) to 4.7 times (160 PNs; 3.5–13.2) larger than other probes, against 1.06 at 512 (Table IV). Scrambling which KCs carry the change within each MBON, which preserves every MBON’s weight-change distribution exactly, left the direction of the MBON current intact (cosine 0.90–0.98) but removed 34–89% of its magnitude for the cued probe (Fig. 4). The retrieval key is thus the identity of the KCs that were active at writing and are re-activated by the cue; it exists only when the KC code is sparse. A repeat with twelve different write pairs gave the same picture: after the KC-identity scramble the cued probe kept a median 16% (160 PNs) and 22% (192) of its current magnitude against 85% at 512, with the direction preserved (cosine 0.92–1.00).

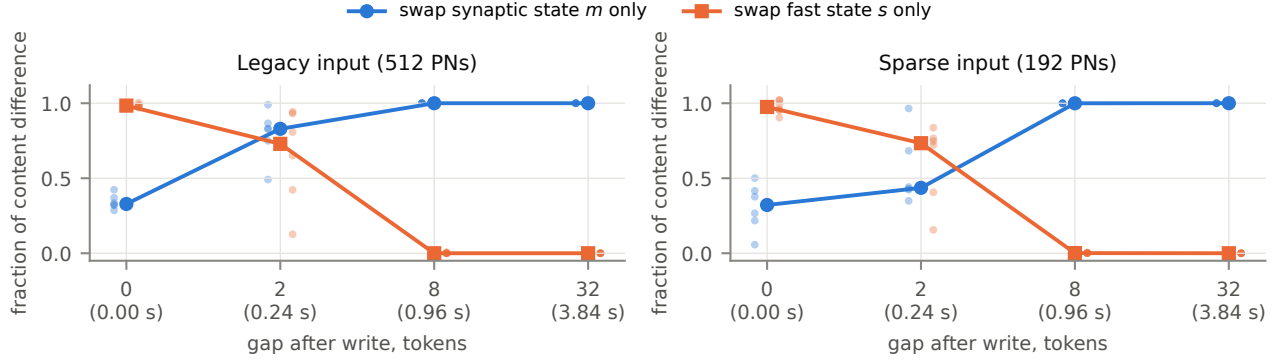


Fig. 1. Transplant decomposition. Response change at the MBON level when only the synaptic state m (circles) or only the fast state s (squares) of the episode is swapped, as a fraction of the natural content difference between two writes. Lines: medians; faint points: individual pair $\times$ probe cells (pairs a/c , b/d ; probes: both written items and a neutral cue).

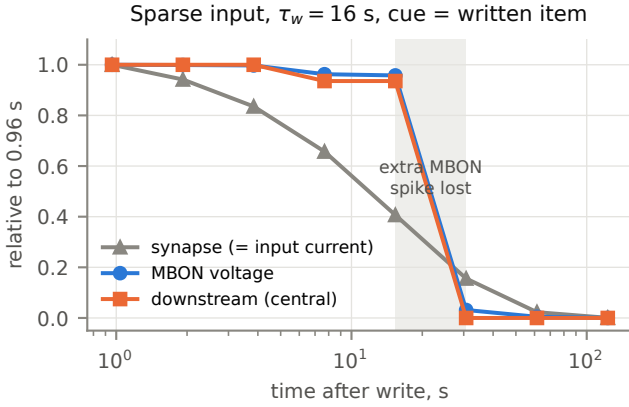


Fig. 2. Retention at sparse input (192 PN, $\tau_w = 16$ s, pair a/c , cue a), each quantity relative to its value at 0.96 s. The synaptic difference equals the cued input current and decays exponentially; the MBON voltage and downstream differences track the presence of one extra MBON spike (shaded: interval in which it is lost).

TABLE IV

ADDRESSING ON THE MBON INPUT CURRENT, SIX WRITE PAIRS. CUE SPECIFICITY: $|I|$ FOR THE WRITTEN ITEM OVER THE MEAN OF ANOTHER ITEM AND A NEUTRAL CUE. SCRAMBLES PRESERVE THE MULTISET OF Δw EXACTLY.

	$n_{PN} = 160$	192	512
Cue specificity (median)	4.7	3.6	1.06
KC-identity scramble, magnitude	0.11–0.59	0.13–0.66	0.82–0.88
KC-identity scramble, cosine	0.90–0.97	0.94–0.98	0.99–1.00
MBON-routing scramble, magnitude	0.73–0.77	0.71–0.76	0.38–0.74
MBON-routing scramble, cosine	0.72–0.75	0.72–0.75	0.44–0.74

E. The cue needs only sparseness; the output identity needs the real wiring

Replacing the PN $\rightarrow$ KC wiring by degree-preserving or uniform random wiring reproduced the KC sparseness, symbol overlap, cue specificity and KC-identity dependence at every sparse operating point, with random or with glomerulus-structured symbol codes. The real wiring did not increase cue

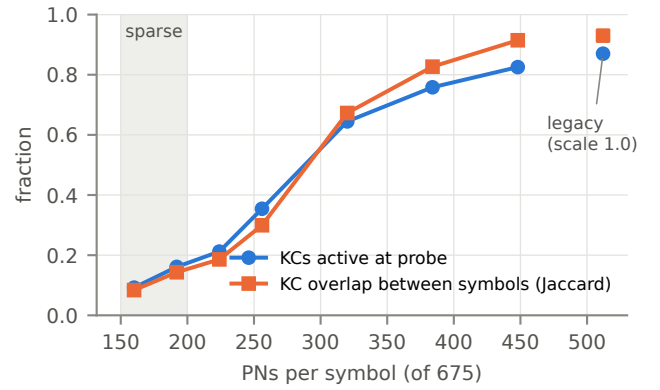


Fig. 3. KC code density against PN code size (drive scale 1.5; the legacy point is scale 1.0). Shaded: the sparse regime used below.

specificity: a pre-stated test (real above the null median in at least 75% of pairs in each of four glomerular-code settings) failed (4/8, 6/8, 6/8, 7/8 pairs), and with random symbol codes the real wiring's median cue specificity over eight items (2.9) lay below that of all four nulls (3.0–6.6).

The identity of the output is a different matter (Table V, Fig. 6). In the connectome, the MBON currents evoked by eight different items have mean pairwise cosine 0.76, with 57–60% of each item's current in five MBONs. Randomising the KC $\rightarrow$ MBON wiring raises the cosine to 0.96–0.98 and spreads the current (top five MBONs: 27–29%); randomising the PN $\rightarrow$ KC wiring raises it to 0.97–0.99. Cue specificity is unchanged by either.

Controls that preserve structure locate what matters. Swapping whole KC output profiles among KCs of the same annotated subtype and the same body side leaves distinctness intact (0.745–0.756 against 0.756–0.764), as does permuting PN inputs among PN of the same type and side (0.72–0.77). The same swaps with the two sides mixed destroy it (0.90–0.95), and so does every earlier control that mixed the sides. Neither individual KC wiring nor any fine matching

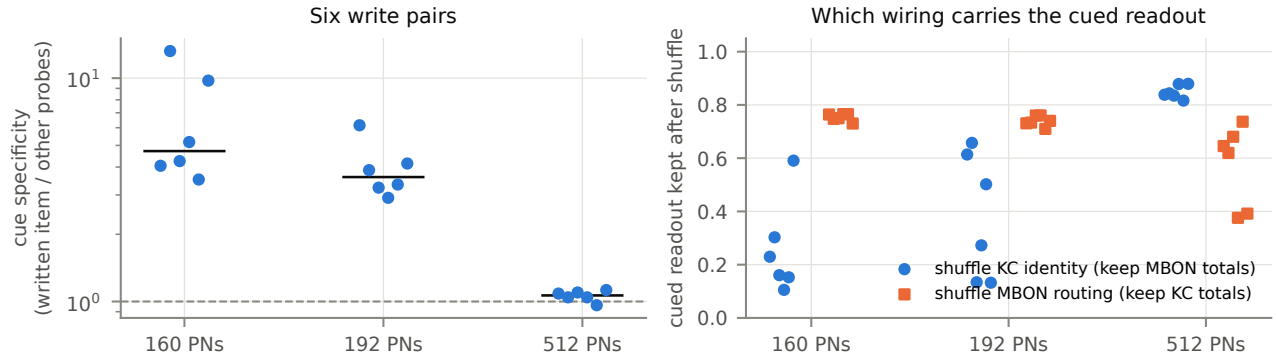


Fig. 4. Addressing on the MBON input current, six write pairs. Left: cue specificity (bars: medians; dashed: no specificity). Right: fraction of the cued readout magnitude kept after shuffling KC identity within each MBON (circles) or MBON routing within each KC (squares); both shuffles preserve the multiset of weight changes exactly.

between a KC's inputs and outputs is required; body side and KC subtype are. With three further shuffle seeds the side-and-subtype swap again kept distinctness (0.741–0.751 at 160 and 192 PNs), while the side-mixed swap (0.90–0.94), KC→MBON randomisation (0.96–0.98) and PN→KC randomisation (0.97–0.99) removed it. A static model that sums the connectome output profiles of each item's written KCs, weighted by their activity, reproduces the simulated distinctness (0.76 and 0.72 against 0.91–0.93 and 0.86–0.87 after within-subtype swapping), so the network dynamics add little to this step.

What differs between items depends on the input code. The side-blind random codes are balanced across the body midline to within a few percent (standard deviation of the left share of PNs across 26 items: 0.03), but the sparse KC threshold amplifies these small differences: the left share of the KCs an item writes ranges from 0.04 to 0.95 (standard deviation 0.23–0.26), and it predicts the left share of the MBON output ($r = 0.96$). Items are then stored largely in one of the two mushroom bodies. At dense input this lateralisation disappears (0.43–0.56). Real odours activate homologous glomeruli on both sides, so we repeated the analysis with bilaterally symmetric glomerular codes (all PNs of a set of PN types). Lateralisation then nearly vanishes (standard deviation 0.06–0.08), distinctness is reduced but remains clearly above the nulls (0.88 against 0.96–0.99 for either randomisation), and it now depends on KC subtype rather than on side: swapping output profiles within subtype with sides mixed keeps it (0.89–0.90, reproduced with three new seeds), while keeping sides and mixing subtypes removes it (0.95–0.97). Items differ in the KC subtypes they recruit (share of the written activity across items: $\alpha\beta$ 0.42 ± 0.15 , γ -main $0.36\text{--}0.38 \pm 0.11$, $\alpha'\beta'$ -ap1 $0.10\text{--}0.12 \pm 0.07\text{--}0.11$), and each subtype projects to its own lobe's MBONs. Subtypes that receive little olfactory input in the fly, γ -dorsal and $\alpha\beta$ -posterior, were almost never recruited ($< 0.3\%$), a sanity check on the PN wiring.

With a single teacher, the dopaminergic gate carries no item identity: across items the per-MBON gate vectors were

identical (cosine 1.000), clamping every item's gate to the across-item mean left distinctness unchanged (0.756, 0.764), and replacing the gate by +1 on every gated MBON, which removes the hand-set valence signs, changed it by at most 0.02.

The subtype-to-MBON map (Fig. 5) is block-structured and follows the lobe organisation: $\alpha\beta$ KCs supply MBON14 and neighbouring α -lobe MBONs, $\alpha\beta$ -posterior KCs supply MBON19, $\alpha'\beta'$ subtypes supply MBON10/13/15/16/17/28, γ -main KCs supply MBON01/05/21 and γ -dorsal KCs MBON27 (connectome annotation names). Under bilateral codes, an item's KC-subtype mixture alone, combined with the mean output profile of each subtype, explains 66% of the item-to-item variation of the output ($R^2 = 0.66$, 26 items).

F. When teachers differ, the teacher chooses the compartment

Real memories are taught by different dopaminergic neurons. We wrote six items with four teachers: all PAM, all PPL, and two halves of the PAM types (PAM01–07, PAM08–15). The teachers produced different gates (cosine between the two PAM halves 0.19; between PAM and PPL -0.10), and the same item written by different teachers produced nearly unrelated outputs (cosine 0.085 between the PAM halves, -0.06 between PAM and PPL). Across the 24 item–teacher combinations, the teacher explained 87% of the variance of the normalised output, the item 5%, and their interaction 7%; within a teacher, items remained as distinct as before (0.89–0.91). Consistent with the compartmental organisation of dopaminergic input [2] and with the cell-type-specific writing rules of individual DAN types in vivo [18], the dopaminergic system therefore decides *where* a memory is stored, and the KC layer decides what is stored there and which cue retrieves it. The sign of the change is not a result: in this model PAM writes potentiate and PPL writes depress (100% of changed synapses in each case) because of the hand-set valence signs, whereas in the fly coincident KC and dopaminergic activity predominantly depresses KC→MBON synapses [1]. Re-running the location results with every DAN depressing its compartment gave the same values (cue specificity 4.71, 3.59

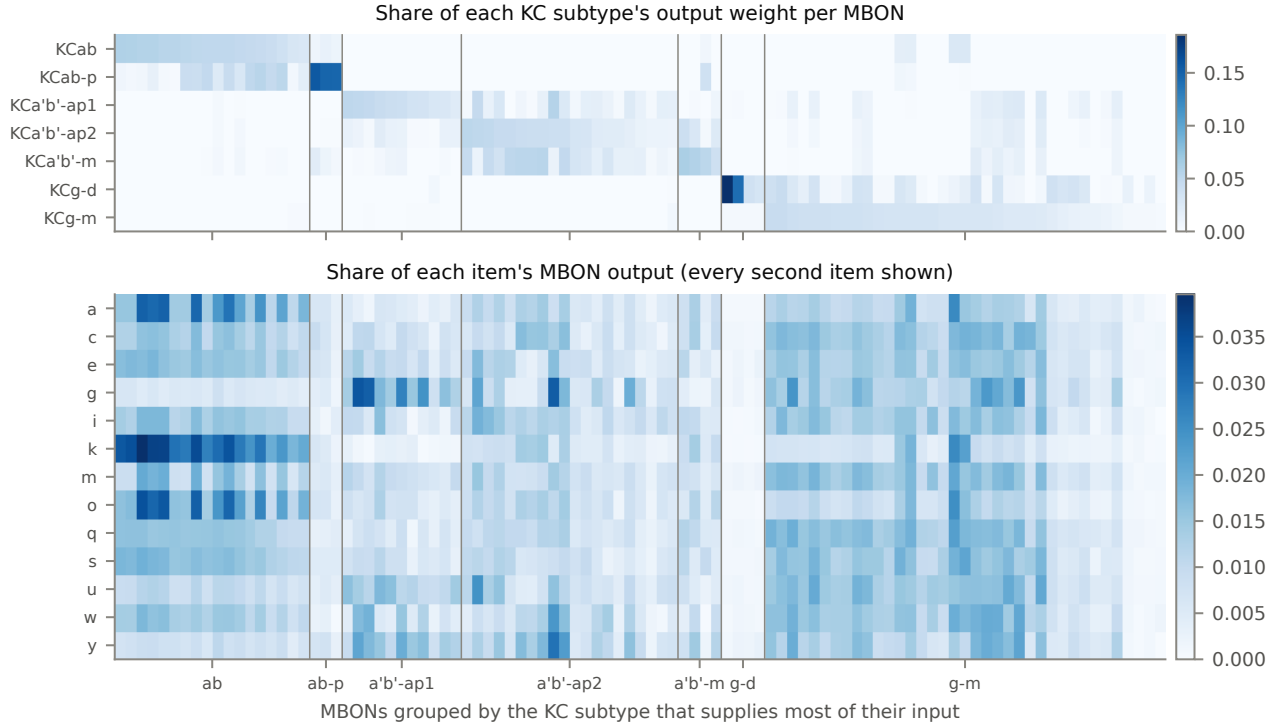


Fig. 5. Output map at 192 PNs with bilateral glomerular codes. Top: share of each KC subtype’s total output weight going to each MBON; MBONs are grouped by the subtype that supplies most of their KC input. Bottom: share of each item’s output (activity-weighted sum of its written KCs’ output profiles) per MBON, for every second of 26 items.

and 1.07; item distinctness 0.885; teacher share 87%). For single-teacher norms and cosines this is largely guaranteed by symmetry, so the check mainly shows that network feedback does not depend on the sign.

G. Interference between memories is additive and set by KC overlap

Writing a second item Y after X changed the cued readout of X by exactly the amount that Y ’s write alone produces when read by cue X (median 0.29 predicted and measured with the same teacher, 0.55 against 0.55 with different teachers, 1.57 against 1.57 at dense input). Interference is linear superposition in the weights, and its size follows the overlap between the KCs Y writes and those cue X activates: 0.10–0.13 at sparse input against 0.86–0.90 at dense input, where Y ’s contamination exceeds X ’s own trace 1.6- to 2.9-fold. With the same teacher, the contamination lies along X ’s own output direction (cosine 0.999); with different teachers it adds a new direction (0.88 sparse, 0.58 dense), because cue X reads Y ’s changes in other compartments through shared KCs. Separate compartments thus do not prevent interference; sparse KC codes do. That sparse, high-dimensional KC codes reduce interference between associations has been shown in an abstract model of the fly circuit [19]; here the same effect appears in the whole-CNS connectome model, where it is exactly linear superposition in the weights and is quantified by the measured KC overlap.

TABLE V
ITEM DISTINCTNESS OF THE MBON OUTPUT UNDER REWIRING, SIDE-BLIND RANDOM CODES (8 ITEMS). MEAN PAIRWISE COSINE BETWEEN ITEMS’ MBON CURRENTS; RANGES OVER NULL DRAWS.

Wiring	$n_{PN} = 160$	192
Connectome	0.756	0.764
Swap profiles, same subtype and side	0.749–0.754	0.745–0.756
Swap profiles, same side only	0.79–0.81	0.82–0.84
Swap profiles, same subtype, sides mixed	0.94–0.95	0.90
KC→MBON shuffle within side	0.80	0.83
KC→MBON shuffle within class, sides mixed	0.94	0.90–0.91
KC→MBON shuffle, all	0.96	0.98
PN→KC shuffle, same PN type and side	0.72	0.77
PN→KC shuffle within side	0.85–0.87	0.85
PN→KC shuffle, same PN type, sides mixed	0.94	0.94
PN→KC shuffle, all, degree-preserving	0.98	0.98–0.99
Dense input (512 PNs): connectome / shuffle	0.91 / 0.95–0.96	

H. Downstream neurons see the memory only through MBON spikes

In the sparse regime the cued MBON response is mostly subthreshold (about one MBON spike per probe), and central targets showed no change at the numerical floor. Across 128 cells (four operating points, four levels of an added MBON depolarising bias, four pairs, two probes), the central response differed from the no-write response in all 31 cells in which the write changed the MBON spike count and in all 4 in which it moved a spike without changing the count, and in none of

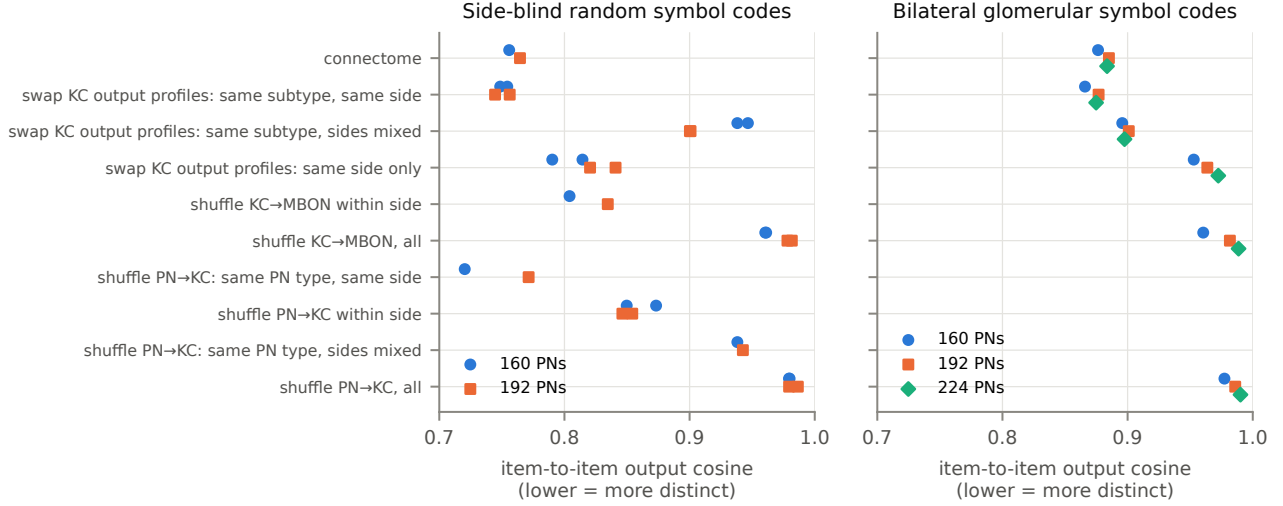


Fig. 6. Item distinctness of the MBON output (mean pairwise cosine between the input currents evoked by eight items; lower = more distinct) in the connectome and under wiring nulls, for side-blind random symbol codes (left) and bilaterally symmetric glomerular codes (right). One point per null draw.

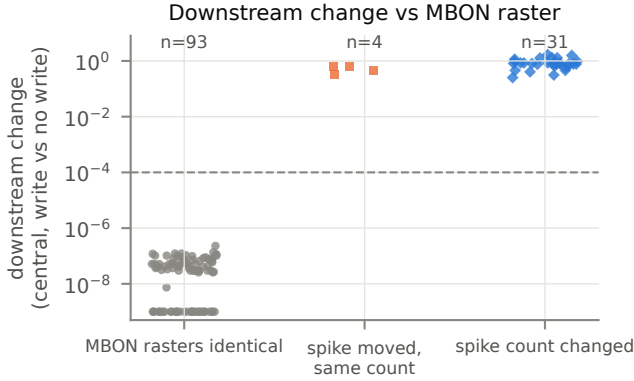


Fig. 7. Downstream (central) response change caused by the write, grouped by whether the write changed the MBON spike raster (128 cells). Dashed: $1e-4$. Exact zeros are drawn at $1e-9$.

the 93 cells with identical MBON rasters (maximum change 2.3×10^{-7}). Raising MBON excitability did not open this gate reliably, because which threshold crossing flips is discrete. The memory is therefore transmitted as a small number of added, removed or shifted MBON spikes, which explains the step-like readout curves (Fig. 7).

IV. CORRECTIONS TO EARLIER ANALYSES OF THIS MODEL

- *Wrong-synapse writes.* Before a simulator fix on 2026-09-17, the masked weight matrix was uploaded with unsorted column indices and re-sorted in place by the first GPU matrix product, after the plasticity module had cached its edge positions. In that state 61,164 of 61,210 cached positions point to a different presynaptic cell (11,130 to non-KCs), and writing back the unchanged weights alone alters the MBON input rows by 63% (L2). The capacity comparison of the earlier draft (frozen 1.17 against plastic

1.16) and the earlier retention curves predate the fix. Re-run after the fix, the frozen value reproduces (1.162–1.186 over three seeds) but plasticity *lowers* activity-based recall (1.126–1.131; paired differences -0.034 to -0.056), and the synaptic state carries history beyond the input window (lags 4–12: 0.11–0.16) that activity does not (0.004–0.028).

- *Input echo.* The capacity task injected the previous three symbols together with the current one; its lag-1 to lag-3 accuracies measure the input window, not network memory.
- *Predictions true by construction.* The reported invariance of the no-write response geometry to plasticity dose and delay holds because the no-write branch performs no write and the network is silent during the delay. The reported rescaling without rotation of the write state across gaps holds because the gap applies a scalar decay.
- *An intermediate claim of this study, withdrawn.* An earlier version of this analysis concluded that output identity required fine, individual-level matching between each KC's inputs and outputs. That followed from controls that all mixed the two hemispheres; side-preserving controls (Table V) show that body side and KC subtype suffice.
- *Dense operating point.* All earlier experiments used 512 of 675 PNs per symbol, where the KC code is dense; the reported probe-fixed direction of the write effect and the near-collinearity of probe responses follow from all probes activating nearly the same KCs.

V. DISCUSSION

That sparse KC codes could provide addressable memory locations was proposed from experiments [5], and that dopaminergic neurons gate plasticity compartment by compartment is part of the known MB architecture [2], [1]. What the model adds is a causal separation of these roles from each

other and from coarse and fine wiring, with the conditions under which each operates.

In this model, a mushroom-body memory has three addresses. The KC layer supplies the retrieval key: when input is sparse, only a cue that re-activates the written KCs retrieves the change, and random PN→KC wiring suffices for this, as expected from random-expansion accounts [6], [9] and from the view of the PN→KC expansion as a locality-sensitive hash [20]. The output identity, in contrast, requires the real wiring, but only its coarse organisation: which hemisphere, and which KC subtype, a KC belongs to on both its input and its output side. Within a subtype and a side, KCs are interchangeable, consistent with random glomerulus-to-KC convergence [6]. An item’s output is the sum of the output profiles of the KCs it recruits, and items differ at the output to the extent that they recruit different mixtures of mushroom bodies and of KC subtypes, which the lobe-specific KC→MBON organisation [2], [8] maps onto different MBONs. In the connectome the mean output profiles of the three KC families ($\alpha\beta$, $\alpha'\beta'$, γ) are nearly orthogonal (cosine 0.01–0.20), as are those of the left and right mushroom bodies (0.00–0.06), so “side and subtype” amounts to “which lobe output channels”: the anatomy defines the channels, and the input code decides how an item is spread over them. How strongly they do so depends on the input: side-blind codes produce strongly lateralised storage, which is unlikely for real odours; bilaterally symmetric codes produce weaker, subtype-based distinctness. Both addresses depend on an operating point with a sparse KC code [4], [5]; at dense input the MB collapses to a set of per-MBON scalars and loses both. A third address is supplied by the dopaminergic system: which DANs teach decides which compartments store the memory, and when teachers differ this dominates the output (87% of its variance). The resulting division of labour is: the DAN chooses the compartment, the KC subtype mixture and body side shape the output within the compartments the DAN allows, and the identity of the written KCs is the key that retrieves it.

Two properties are set by modelling choices rather than by the wiring. Memory lifetime is exactly the prescribed synaptic decay, and the network contributes no maintenance because it is silent between events; claims about forgetting or consolidation require intrinsic activity, feedback plasticity or DAN→MBON dynamics that this model lacks [12], such as the dopamine-mediated interactions between short- and long-term memory units found in vivo [16]. And the readout’s all-or-none character follows from integrate-and-fire transmission: downstream neurons see the memory only when it changes an MBON spike. Our MBONs are point neurons; a biophysical model of MBON- $\alpha 3$ built from its EM architecture shows that dendritic integration of stochastic KC input shapes the MBON response [15], and how graded such a readout is in a compartmental model is open.

Table VI separates what the simulations discovered from what the model builds in. Four of the seven findings follow, in whole or in part, from the model’s rules or from our readout metric; their contribution is a quantification, not a mechanism.

The two findings that depend on the wiring and the operating point, the KC retrieval key and the side-and-subtype output map, were reproduced across items, shuffle seeds, code sizes and code types; we also varied gain (1.2, 1.8; reference 1.5), tonic drive (0.03, 0.08; reference 0.05) and added membrane noise (2 and 5 Hz), rerunning the real wiring in each job (the reference reproduced 0.756 and 0.885 exactly). The side-and-subtype swap kept output distinctness within 0.016 of the real wiring in all 14 variant-and-code combinations, while randomising KC→MBON wiring removed it in all of them. The retrieval key changed strongly, but through one variable only: at 160 PNs cue specificity fell as the KC overlap between items rose (Spearman $\rho = -0.96$ over the seven settings) (overlap 0.05, 0.08, 0.17, 0.46, 0.58, 0.60, 0.94; specificity 9.9, 9.3, 4.7, 2.8, 2.9, 2.0, 1.95), whichever parameter raised it. Higher gain, stronger tonic drive and noise all densify the KC code and weaken the key; the dependence on sparseness, not on a particular parameter setting, is the robust result (Fig. 8).

In the fly, KC sparseness depends on feedback inhibition from the APL neuron [5]. In the model it does not. The connectome’s KC→APL synapses sum to a weight of 1.8 (APL→KC: -378), and removing APL→KC inhibition left every number at 160 PNs unchanged (it densified the already dense 512-PN code slightly: probe KCs 3,446 to 3,757, specificity 1.06 to 1.00). Scaling KC→APL by 100 or 1,000 sparsened the KC code (probe KCs 437 to 169 and 135) and sharpened the key (specificity 4.7 to 10.5 and 8.9), as expected from [5], but did not rescue dense input (specificity 1.24; the side-and-subtype swap still kept the output map). Sparseness in this model is therefore set by the input, not by the APL loop: a point-neuron APL with synapse-count weights is too weak, and the non-spiking, local inhibition of the real APL is not captured.

Finally, to rule out another bookkeeping error of the kind described in Section IV, we recomputed the MBON input current in a second implementation that reads weight changes directly from the live connectivity matrix, selecting KC→MBON entries by cell type rather than through the plasticity module’s slot tables. For six pairs at 160 and 512 PNs the two implementations agreed exactly in all 36 cells.

Predictions: The model makes four predictions that experiments in the fly could test. (1) Retrieval specificity should fall smoothly as the overlap between the KC sets of two stimuli rises, whatever raises it (stronger PN drive, weaker inhibition, more similar odours); the model gives the shape of this relation (Fig. 8). Impaired discrimination after blocking the inhibitory APL neuron [5] is the qualitative case. (2) Silencing one KC family after training should remove the part of a memory expressed by that family’s lobe MBONs and leave the rest, and the effect of a memory on MBON output should depend on how an odour’s KC response is divided between families, not on which individual KCs respond. (3) A stimulus delivered to one side only should be stored and retrieved largely in one mushroom body. (4) Training with two different DAN groups should change largely non-overlapping sets of MBONs; this is consistent with compartment-specific memories [2], [18] and

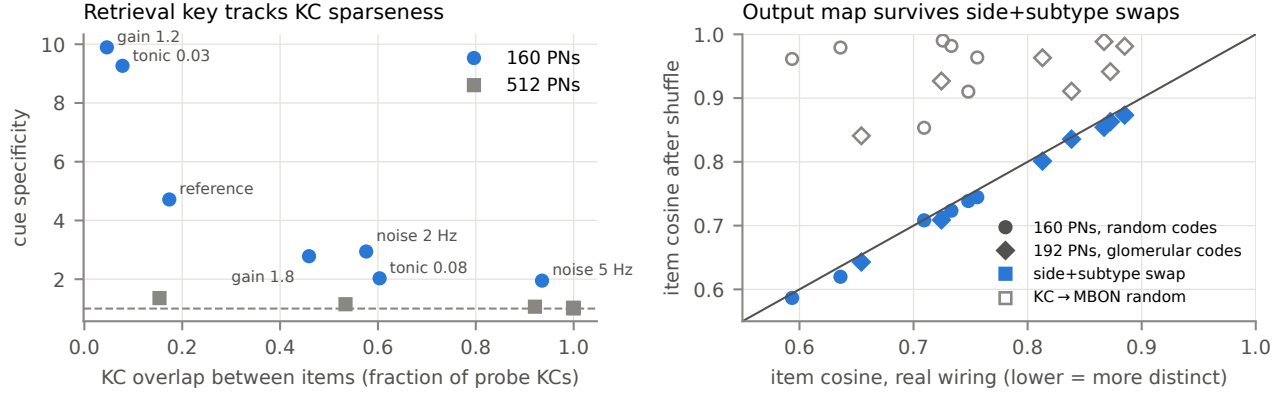


Fig. 8. Sensitivity to gain, tonic drive and membrane noise. Left: cue specificity of the written item (median over six write pairs) against the KC overlap between items, for seven parameter settings; dashed: no specificity. Right: mean pairwise cosine of item outputs with the real wiring against the same after swapping KC output profiles within subtype and side (filled) or randomising KC→MBON wiring (open), for the seven settings and two input codes; points on the diagonal keep the output map.

TABLE VI
STATUS OF THE MAIN FINDINGS.

Finding	Status
Memory on weights after ~ 1 s	Time scale set by the 100 ms membrane constant; the transplant test is the evidence
Lifetime = prescribed decay	By construction; shows only that no other forgetting emerges
KC identity is the retrieval key	Emergent; strength set by KC sparseness across gain, tonic and noise settings (12 pairs, two sparse points)
Side + subtype set output identity	Emergent from the wiring; amounts to the lobe/side output channels (several shuffle seeds, two code sizes and types, seven parameter settings)
DAN chooses the compartment (87%)	Largely by construction (rule applies DAN signal via DAN→MBON edges); the connectome sets how little teachers overlap
Interference additive, set by overlap	Additivity by construction of the current metric; overlap dependence expected from theory [20], [19]
Downstream sees memory only via spikes	By construction of spike-based synapses; the rarity of flips is the finding

is the least novel of the four.

Limitations: One connectome and one reconstruction; leaky integrate-and-fire neurons with hand-set gain, tonic drive and zero noise (the two emergent findings were checked at two other values of each and with 2 and 5 Hz noise, other results only at the reference); an APL feedback loop too weak to sparsen the KC code, so sparseness comes from the input alone; a hand-set plasticity rule and valence signs; symbol codes that are not odour responses; operating points chosen by us; write pairs and items within one connectome as the only replication; input current as a linear proxy that ignores spiking; the sparse regime’s memory is largely subthreshold at the MBON; symbol codes are either side-blind or built from whole PN types, and neither is an odour response; teachers are artificial pulses on whole DAN groups; the default sign of plasticity (PAM potentiates, PPL depresses) is a model

assumption opposite to the predominantly depressing effect of DANs in the fly; location results were unchanged under a depression-only rule, but the sign of stored changes is not interpretable. None of the results is a claim about memory in the fly.

VI. CONCLUSION IN PLAIN TERMS

Think of the mushroom body as a filing system with many drawers. When the fly is taught something, a dopamine signal decides which drawers are opened: a reward-like signal and a punishment-like signal open different drawers, and in our model this choice explains most of where a memory ends up. What is written is a small change in the strength of connections from Kenyon cells, the neurons that represent “what the fly is sensing”, onto the output neurons of the open drawers. To read the memory back, the same small group of Kenyon cells has to become active again. This only works when each stimulus activates few Kenyon cells, as in the real fly; when too many are active, every stimulus looks the same to the output neurons and the memory can no longer tell stimuli apart. Finally, the shape of the output depends on coarse anatomy, on which side of the brain and which family of Kenyon cells stored the memory, rather than on the exact wiring of individual cells.

Two cautions matter as much as the findings. Some results follow directly from how the model was built, such as how long a memory lasts; we list these separately (Table VI). And all of this happens in a simplified simulation of one reconstructed brain. The results are predictions about the fly, not observations of it.

USE OF AI TOOLS

This is independent research by a high-school student, carried out with extensive use of AI systems. Multiple AI agents, run in parallel and in sequence (Anthropic Claude, OpenAI Codex, and DeepSeek via OpenCode), wrote and ran most of the simulation, analysis and figure code, executed the experiments, performed adversarial audits and independent

re-checks of each other’s results, and drafted and revised this manuscript. The author defined the research question, redirected the project from memory capacity to mechanism, decided which experiments to run and which claims to retract, and is responsible for the content. Several errors in earlier AI-generated analyses (Section IV) were found by later AI audits requested by the author; all numbers were traced to output files, and the audit trail is part of the public repository.

DATA AND CODE

All scripts (`ml_core.py` for the snapshot interface and one script per experiment), their JSON outputs and a result register that maps each number above to its output file are available at <https://github.com/LxCenady/Fly-Is-All-You-Need>.

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